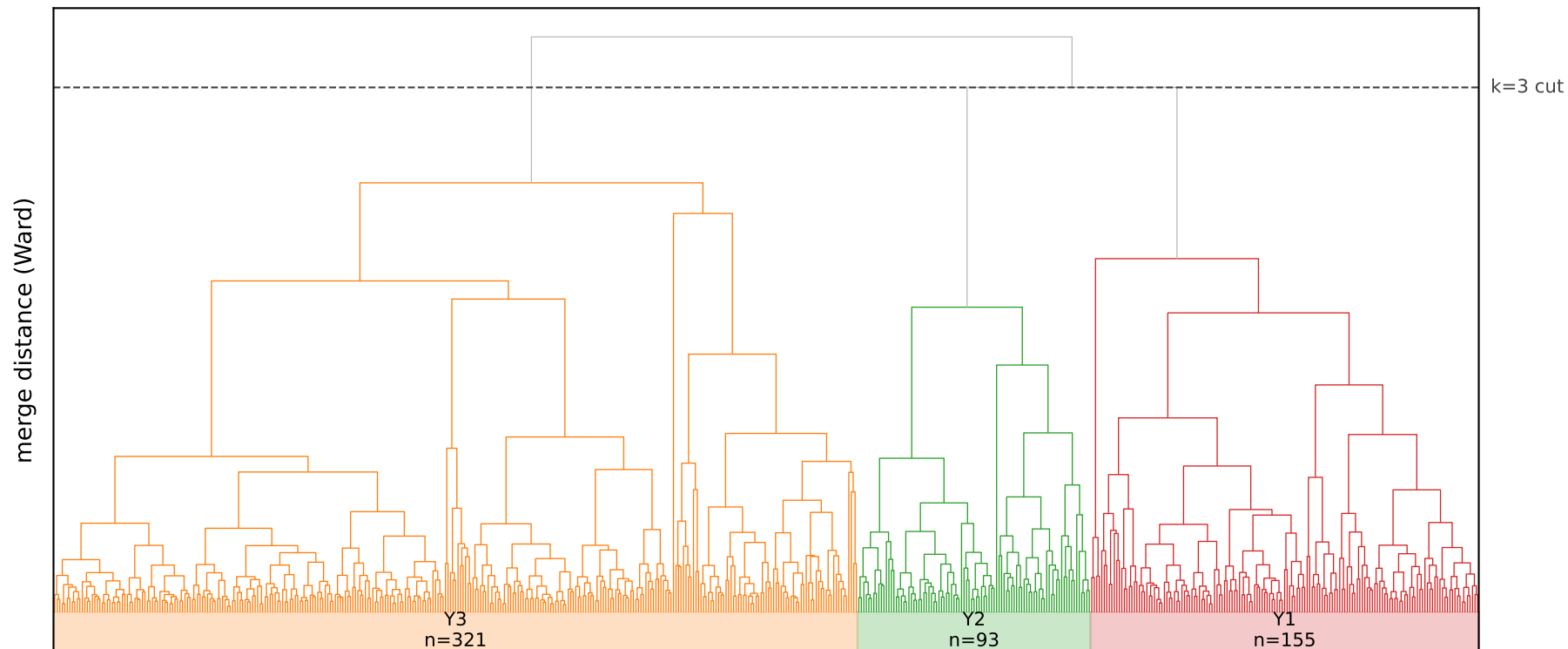


YEAR behavior: Ward dendrogram, coloured at the k=3 cut (n=569)

each cluster is described below by its feature signature; the up/down value is how far its students sit from the whole-group average, in standard deviations



Y3 (n=321)

- ▲ questions attempted +0.4
 - ▲ active weeks +0.4
 - ▲ code lines +0.4
 - ▼ quick resubmissions -0.4
- exam grade mean 42 · med 35

Y2 (n=93)

- ▲ quick resubmissions +1.9
 - ▲ attempts per question +0.9
 - ▼ tries that improved -0.6
 - ▼ gap between tries -0.6
- exam grade mean 35 · med 28

Y1 (n=155)

- ▼ questions attempted -0.9
 - ▼ active weeks -0.9
 - ▼ attempts per question -0.6
 - ▼ code lines -0.6
- exam grade mean 33 · med 20

Clusters are formed from the behavior features only; the exam grade (mean final score over the 6 exam questions, 0 to 100) is shown only to relate the groups to the outcome.